# Module 6

## Implementing Network Services

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# Learning Objectives

- Compare and contrast transport protocols.
- Use command line tools to scan network ports.
- Explain the use of network addressing services.
- Explain the use of name resolution services.
- Configure and troubleshoot DHCP and DNS services.

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# Lesson [6.1]

## Transport & Application Layer Protocols

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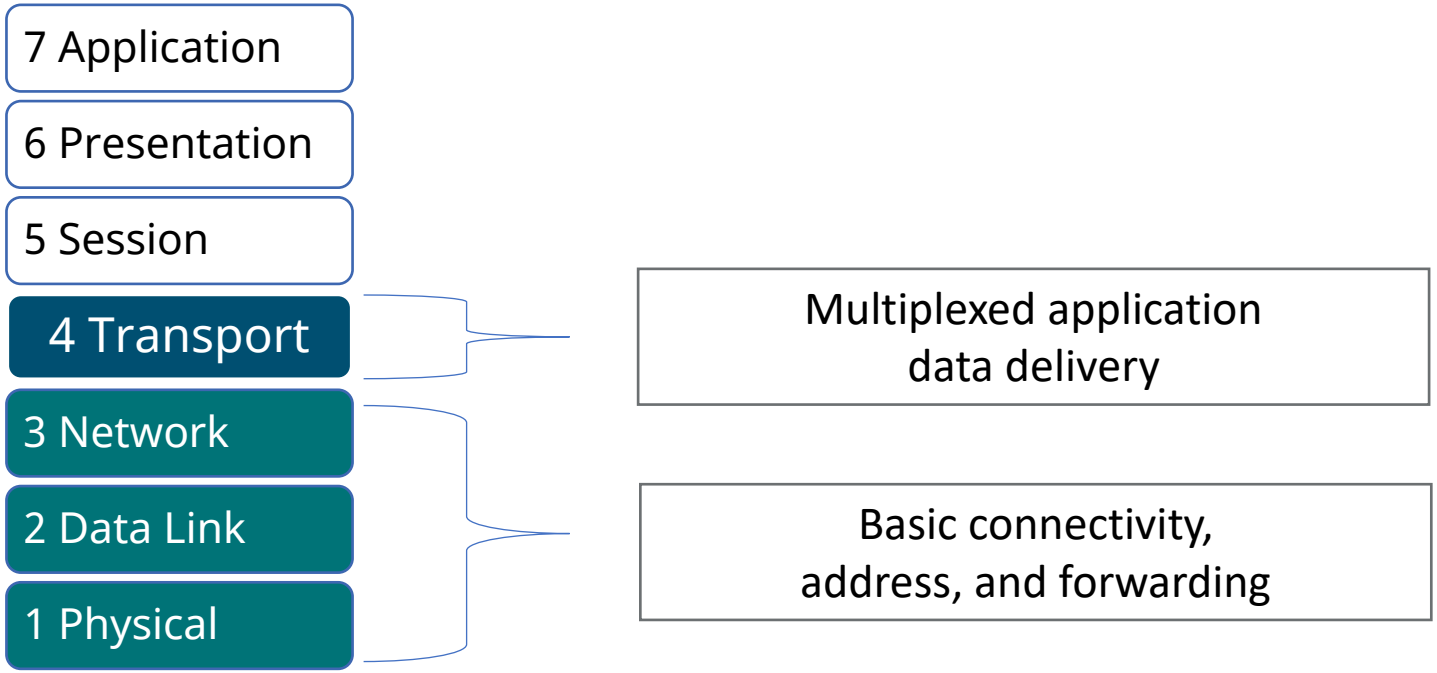
# Transport Layer

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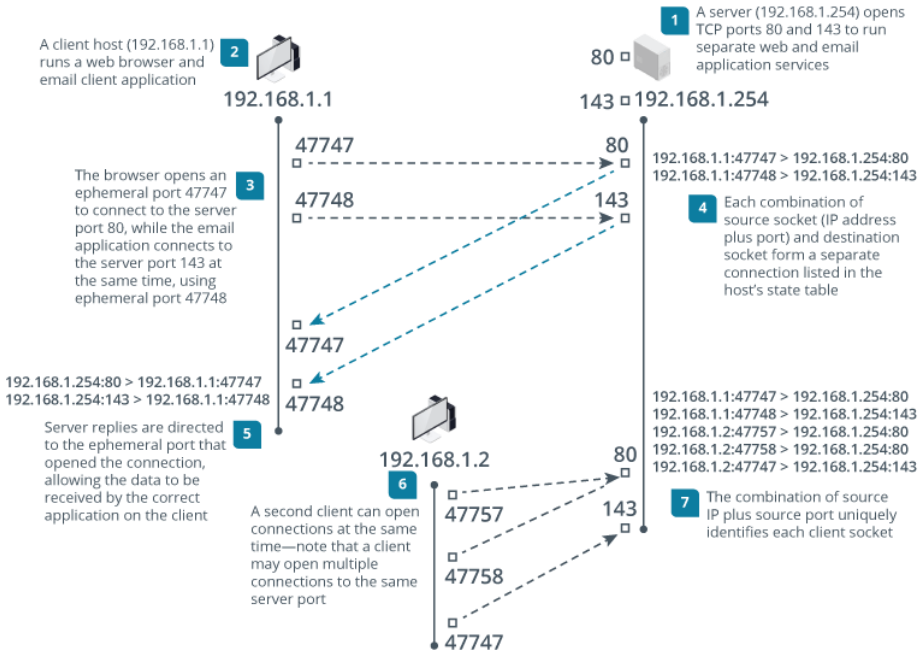
# Transport Layer Ports and Connections

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# Transmission Control Protocol (TCP)

TCP helps computers send information to each other in a reliable way.

For reliability, TCP requires numerous header fields to maintain state information.

- Source port
- Destination port
- Sequence number
- Ack number
- Data length
- Flags
- Window
- Checksum
- Urgent Pointer
- Options

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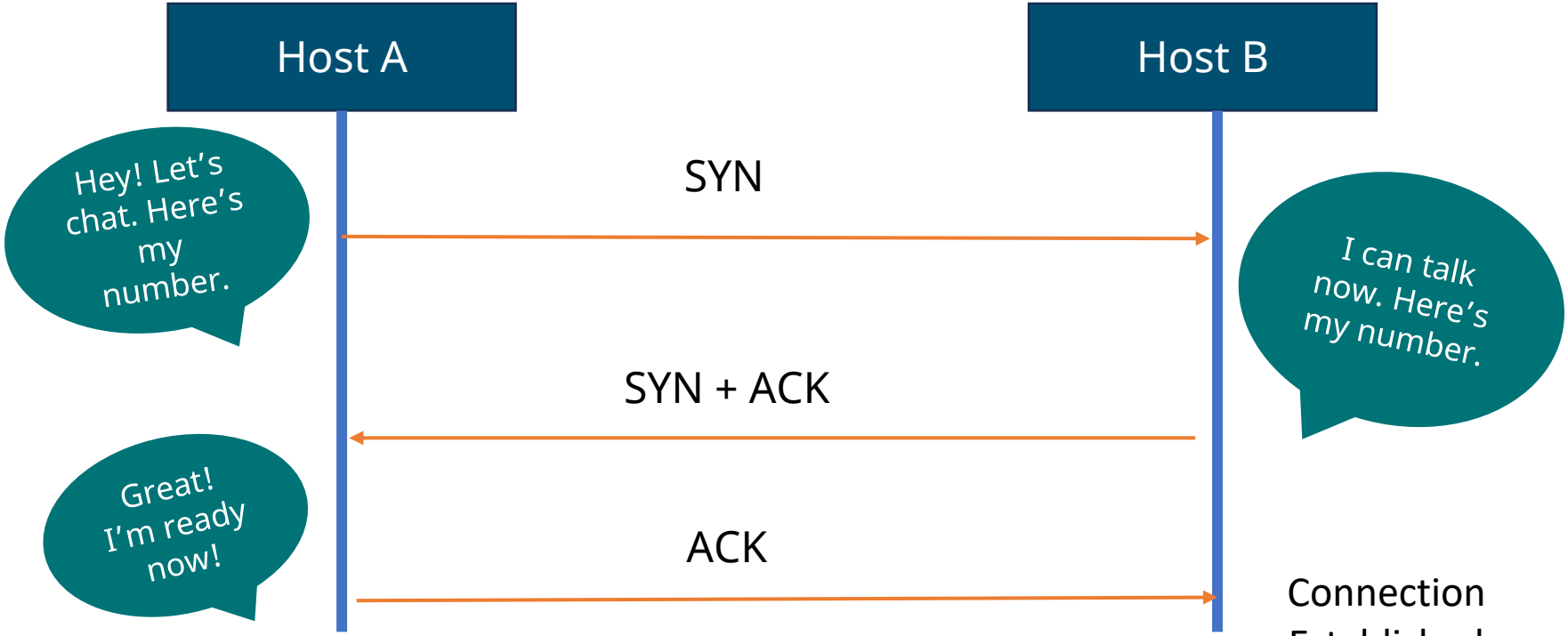
# TCP Handshake

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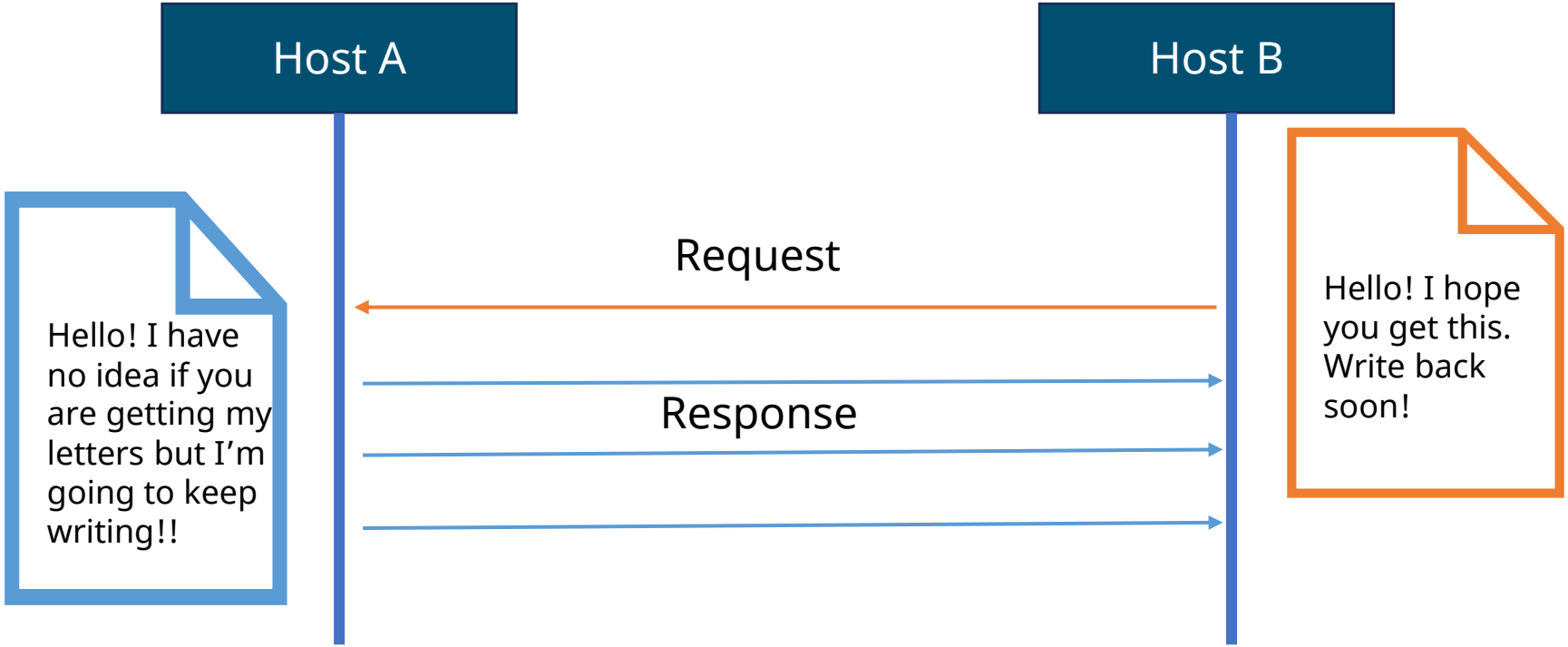
# UDP Exchange

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# User Datagram Protocol (UDP)

UDP helps computers send information to each other in a non-guaranteed way.

- Source port
- Destination port
- Message length
- Checksum

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# Activity: Run netstat

1. Open the command prompt or Terminal.
2. Run **netstat** command to view all active connections and their respective ports.
3. Use additional options like **–a** to show all connections or **–b** to display the application making the connection.

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# Lesson 6.2

## DHCP

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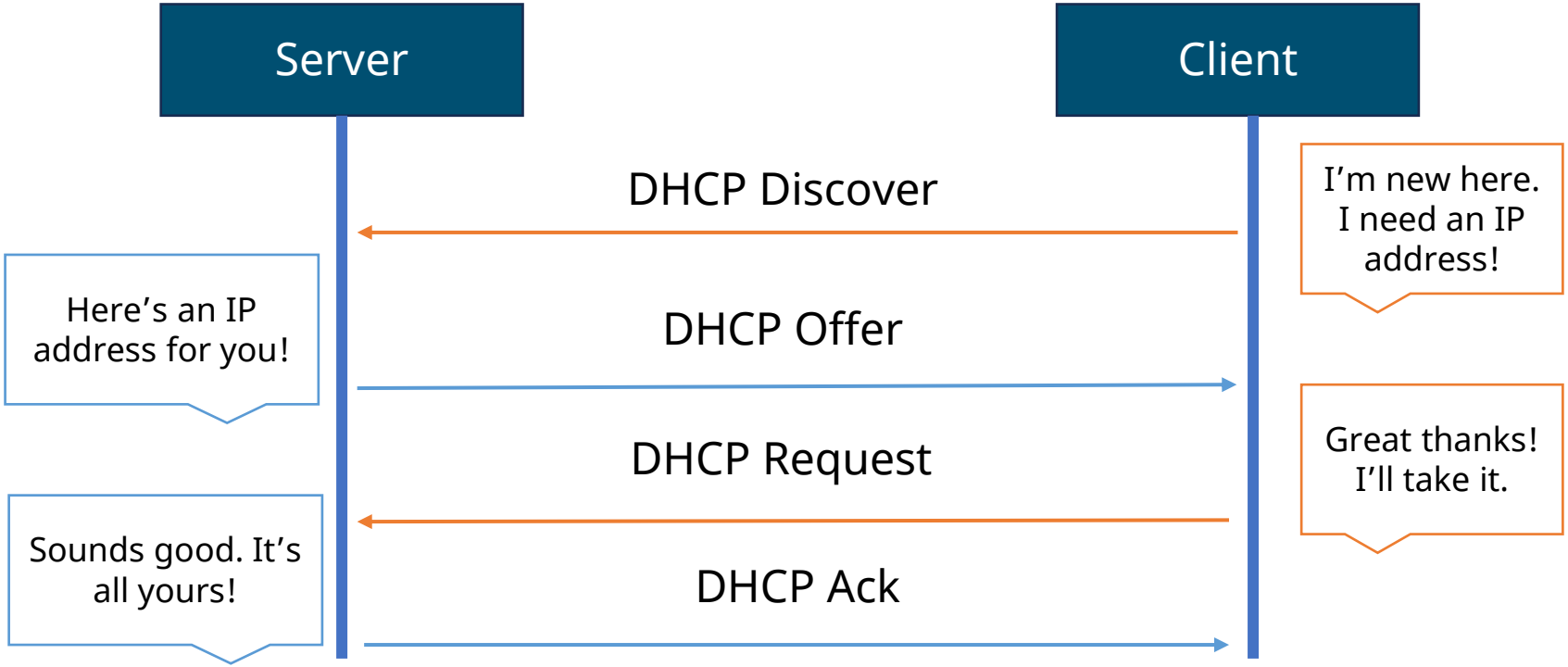
# DHCP Process

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# DHCP Server Configuration

**A DHCP server must be**

- Assigned a static IP address
- Configured with a range of IP addresses/subnet masks (scope)

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# DHCP Reservations and Exclusions

| DHCP Reservations | DHCP Exclusions |
| --- | --- |
| • Ensures a device always receives the same IP address<br>• Tied to a device's MAC address<br>• Ideal for devices that need a consistent IP address | • Prevents a range of IP addresses from being assigned<br>• A range of IP addresses is set aside and not distributed<br>• Ideal for addresses statically assigned to devices |

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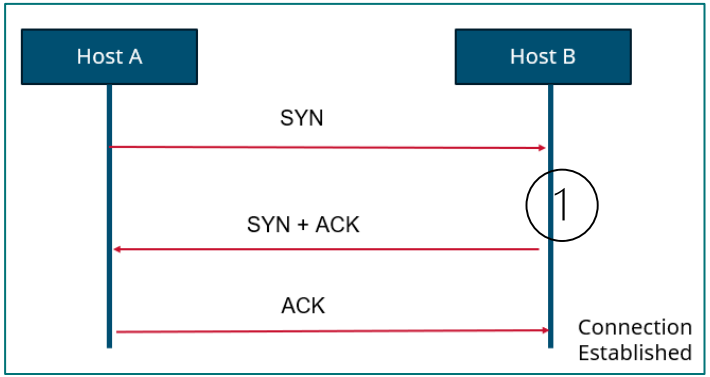
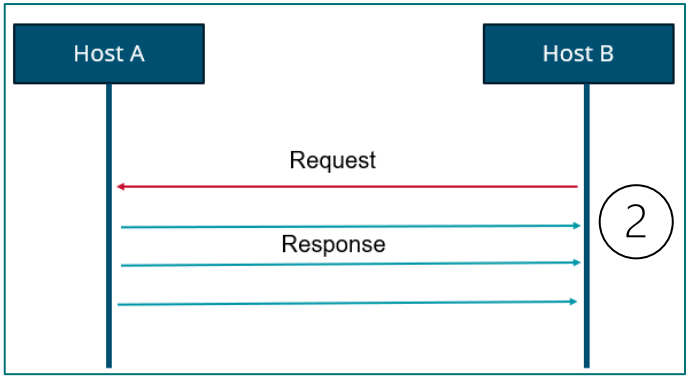
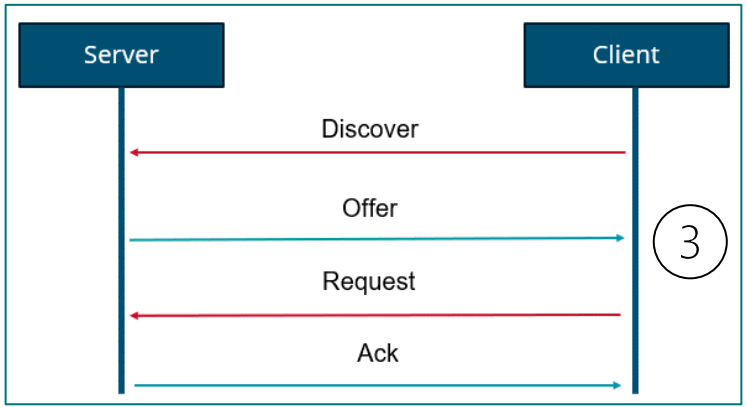
# Activity: Identify the Protocol

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# Lesson 6.3

## APIPA and SLAAC

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# APIPA

## Automatic Private IP Addressing

- Devices can self-assign an IP address when DHCP servers unavailable
- Only for local network use

## How APIPA Works on a Device

- Assigns itself an address between 169.254.0.1 and 169.254.254.255
- Uses ARP to ensure address is unique

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# IPv6 Autoconfiguration

## IPv6 Autoconfiguration Description

- Allows devices on an IPv6 network to configure their own IP addresses without DHCP

## Types

- Stateless Address Autoconfiguration (SLAAC)
- Stateful Address Autoconfiguration (DHCPv6)

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# SLAAC

## Function

- Link-local address configuration
- EUI-64 interface identifier
- Duplicate Address Detection (DAD)

## Advantages

- Simplified network configuration
- No need for a central server

## Limitations

- Does not provide DNS server addresses
- Limited adoption in certain environments

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# DHCPv6

## Function

- DHCPv6 request and response
- Address assignment and management
- Lease time and renewal

## Advantages

- Centralized control
- Extensive network configuration options

## Limitations

- Requires dedicates server setup
- More complex than SLAAC

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# Activity: Think About It

The university's IT department is facing a challenge with the deployment of IPv6 across their diverse campus network.

With a mix of student devices, administrative workstations, and IoT-enabled classroom technology, the team needs an autoconfiguration protocol that can handle dynamic environments and provide ease of management.

**Which protocol should they use?**

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# Lesson 6.4

## DHCP Relay and Troubleshooting

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# DHCP Relay

## DHCP Relay Agent

- Acts as a network bridge
- Receives DHCP broadcast messages from clients and forwards them to DHCP server
- Enables clients to obtain IP address from remote DHCP server

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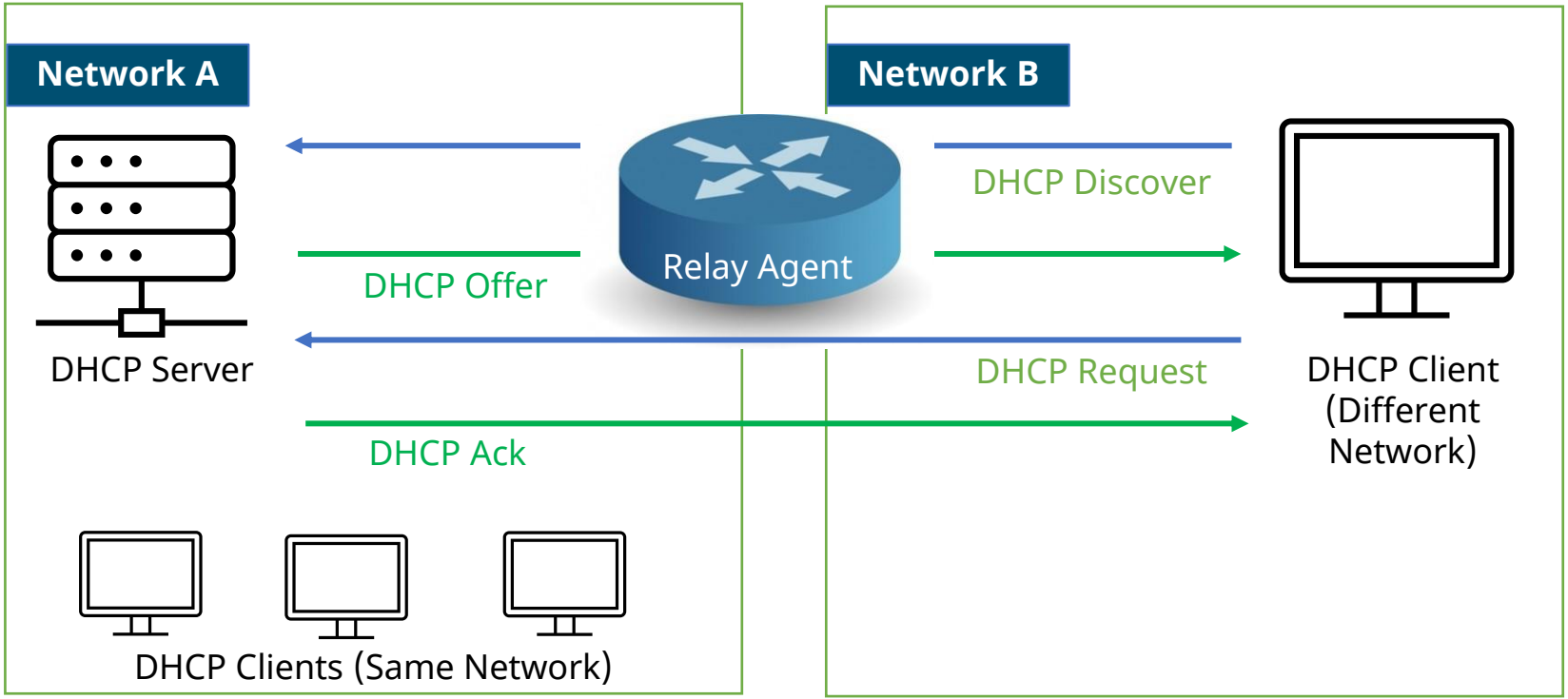
# DHCP Relay

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# IP Helper

## IP Helper

- Cisco network device feature
- Forwards DHCP and other broadcast traffic across different subnets

## Functionality

- Translates broadcasts to unicast to cross subnet boundaries
- Sends DHCP requests from clients to a server in a different subnet

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# Activity: Troubleshooting DHCP

Client is unable to reach external networks

Client received address already in use

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# Lesson 6.5

## Domain Naming System

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# Think About It: DNS

What is DNS
(Domain Name System)?

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# Domain Name Service (DNS)

## Host Names

- Local identifiers
- Assigned by administrator
- Human-friendly names
- Unique on the local network

## Domain Names

- Global identifiers
- Hierarchical structure
- Avoids duplications on the internet
- Registered with a registrar

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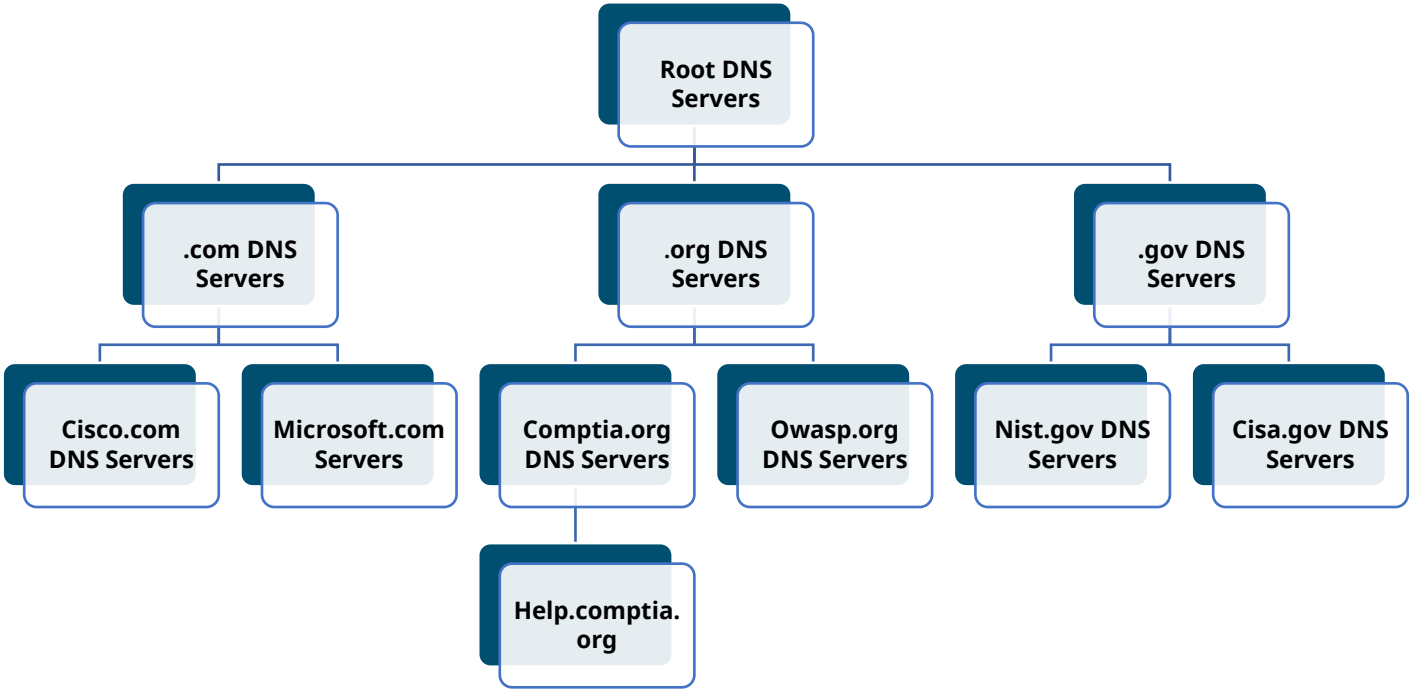
# DNS Hierarchy




Root Domain

Top Level Domain

Domain

Sub-Domain

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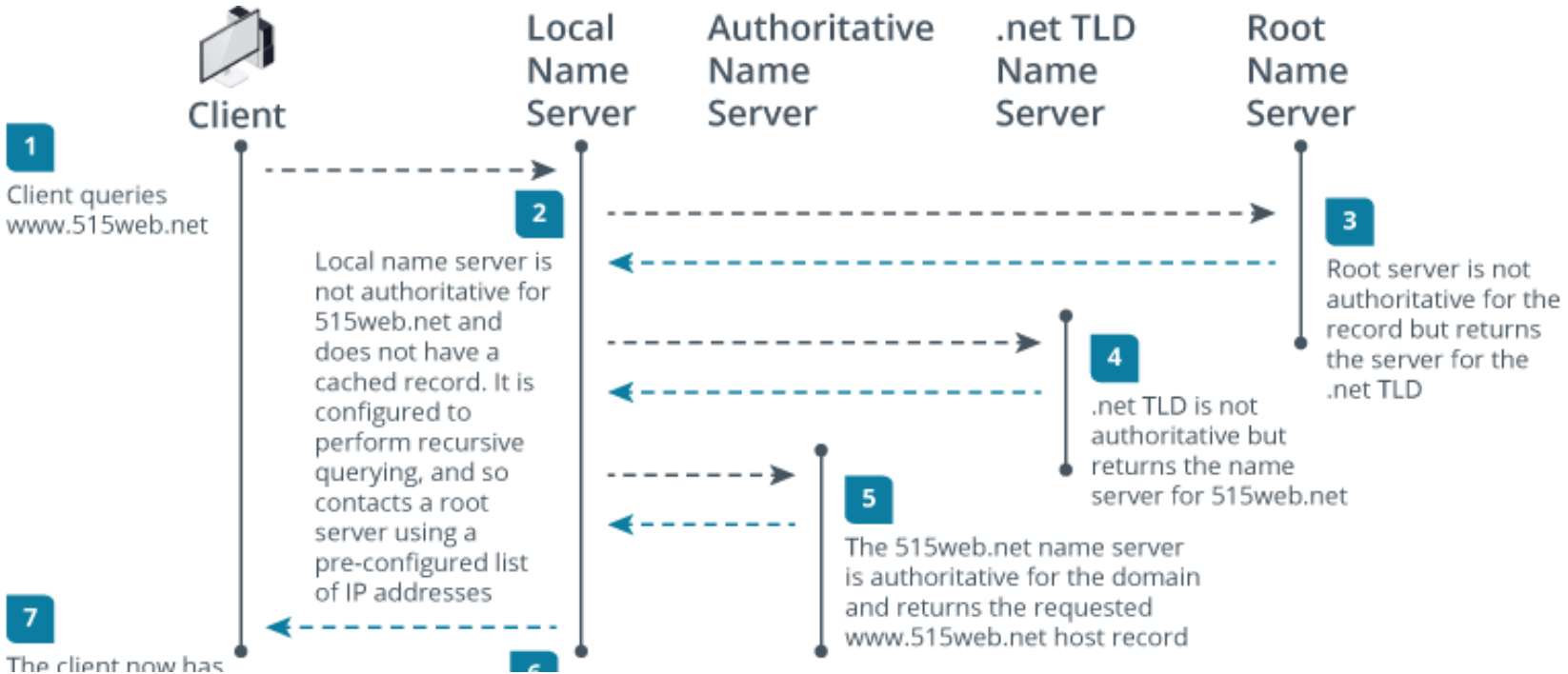
# Name Resolution using DNS

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# Resource Records

DNS name servers maintain the DNS namespace in zones.

A single zone namespace may host records for multiple domains.

A DNS zone will contain numerous resource records.

Resource records help to resolve queries for names and services into IP addresses.

There are numerous types of resource records.

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# Host Record Types

| SOA Records | Host Address Records | Mail Exchange Records | Service Records | Text Records | Pointer Records |
|---|---|---|---|---|---|
| Indicate authority for a DNS zone | Map domain names to IPv4 addresses | Specify mail servers responsible for a domain | Identify service locations within a domain | Store text information for external sources | Used for reverse DNS lookup |
| Contain admin contact, refresh rates, and versioning information | Needed for DNS to translate domain names into IP addresses | Prioritize mail delivery | Include port numbers and priority for services | Used for SPF and DKIM records to reduce spam | Map IP addresses back to domain names |

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# DNS Server Configuration

## Primary DNS Servers

- Translate domain names into IP addresses
- Act as the first point of contact for DNS queries

## Secondary DNS Servers

- Provide redundancy and load balancing for DNS queries
- Serve as backup servers if the primary server fails

## Cache-only DNS servers

- Cache DNS resolutions to speed up response times
- Handle repetitive queries to reduce load on primary/secondary servers

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# Internal vs. External DNS

**External DNS Servers:**

- Public access: Resolves internet domain names for users
- External management: Managed by providers, not local network
- Public focus: Primarily handles public domain to IP resolution

**Internal DNS Servers:**

- Private network: Resolves internal domain names
- Internal control: Managed by IT for local network
- Security focus: Enhances network security by isolating traffic

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# DNS Security

| Threat | Solution |
|---|---|
| DNS spoofing and poisoning | DNSSEC (DNS Security Extensions) |
| Eavesdropping and interception attacks | DNS over TLS (DoT) |
| Interception attacks and tampering | DNS over HTTPS (DoH) |
| DNS tunneling and advanced attacks | Advanced DNS Firewall |

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# Activity: Two Truths and a Lie

Host names help to avoid duplications on the Internet

Resource records help resolve queries for names and services into IP addresses

Pointer records map IP addresses back to domain names

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# Lesson 6.6

## DNS Troubleshooting

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# Client DNS Issues

## Symptoms of DNS Issues

- Timeouts and errors
- Partial DNS failures
- Page loading issues
- Delays in accessing websites

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# Name Resolution Issues

## Name Resolution Methods

- Check local name caches
- Check HOSTS
- Verify DNS records using **nslookup** or **dig**

# Activity: Run nslookup and dig

Windows Machines

- Open command prompt or Terminal
- Type **ns lookup comptia.org**

Linux Machines

- Open Terminal
- Type **dig comptia.org**

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# Summary

- **TCP (Transport)**: Reliable, connection-oriented, and error-checked data delivery (acknowledgments)
- **DHCP (Automatic IP)**: Assigns IP addresses, subnet masks, and optional parameters (gateway, DNS).
- **Relay Agents (DHCP)**: Forward DHCP traffic between subnets
- **DNS (Global Hierarchy)**: Translates domain names to IP addresses.

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